Lecture 1: A Review of Limits

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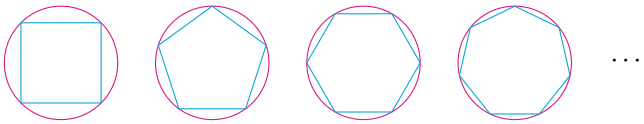
The origins of calculus can be traced back to problems investigated by the ancient Greeks, more than two thousand years ago.

The Area Problem. With basic tools of measurement, the ancient Greeks were able to compute *areas* of polygons. The task of finding the area of a circle, however, was much more challenging. The approach the Greeks used in determining the area of the circle was to use what they already knew about polygons: by inscribing regular polygons of known area into the circle, and increasing the number of sides of these polygons, they could begin to approximate the area of the circle.

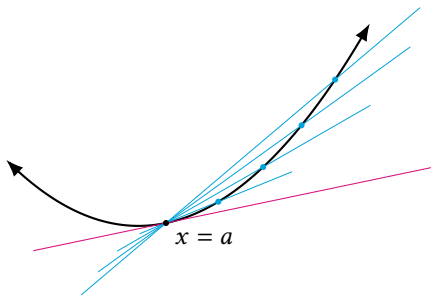


Let A_n be the area of the regular polygon with n sides inscribed in the circle. As n increases, the value of A_n gets closer to the area of the circle. Thus, the area of the circle is the *limit* of A_n as n gets infinitely large, and we write

$$\text{area of circle} = \lim_{n\to\infty} A_n.$$

The Tangent Problem. Two points define a line. We might, however, be interested in finding the *tangent* line to a curve at a point: this line exists but is only determined by one point. If we wanted to describe such a line with an equation, we would need to know its slope (or *gradient*), as well a point that lies on the line. Since we already have the coordinates of such a point, all that is left to determine is the slope.

Suppose wished to compute the slope of the tangent line to the curve $y = f(x)$ at the point where $x = a$.



One way of determining the slope is to draw lines whose slopes we can determine, connecting the point at $x = a$ to other points on the curve. Recall that the slope of a line is given by

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{f(x) - f(a)}{x - a}.$$

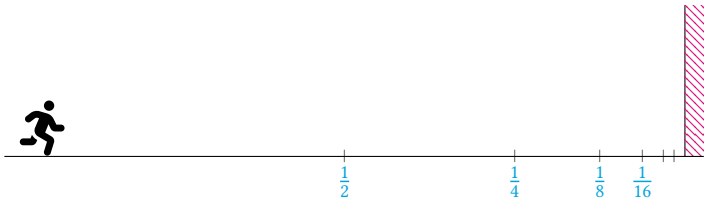
By drawing lines to points that get closer and closer to $x = a$, we can thus approximate the slope of the tangent line with

$$\text{slope of the tangent line} = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}.$$

Zeno's Paradox. The Greek philosopher Zeno of Elea (5th Century BC) posed the following paradox, as recounted by Aristotle in *Physics*.

> That which is in locomotion must arrive at the half-way stage before it arrives at the goal.

Someone standing in a room, for instance, cannot possibly reach the wall on the other end: to do so, they would first have to go half the distance, then half the remaining distance, and then half again of what remains, in an infinite process that never ends.



In reality, we know that the person is indeed able to reach the wall on the other end: this suggests that the total distance, which is finite, can be expressed as the sum of infinitely many smaller distances:

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots+\frac{1}{2^n}+\cdots.$$

Limits

We recall the definition of a **limit**.

Definition. Suppose that a function $f(x)$ is defined when x is near some number a. We write

$$L=\lim_{x\to a} f(x),$$

and we say that

$$L \text{ is the } \textbf{limit} \text{ of } f(x) \text{ as } x \text{ approaches } a$$

if we can make the values of $f(x)$ as close to L as we would like by taking x sufficiently close to a.

This just means that the value of $f(x)$ approaches L as x approaches a. Note that such a limit only depends on the values of $f(x)$ when x is near a, and does **not** depend on the the value of $f(x)$ when $x=a$.

Example. Let us find

$$\lim_{x\to 1}\frac{x^2-1}{x-1}.$$

We can do this graphically: let

$$f(x)=\frac{x^2-1}{x-1}.$$

By plotting* the graph of $f(x)$, we find that as the value of x approaches 1, the value of $f(x)$ approaches 2. So, we can conclude that

$$\lim_{x\to 1}\frac{x^2-1}{x-1}=2.$$

* You can try this out desmos.com/calculator.

A small warning: $f(1)$ is *undefined*, since

$$f(1) = \frac{1^2 - 1}{1 - 1} = \frac{0}{0}.$$

We can also compute the limit algebraically: observe that we can *factor* the expression in the numerator.

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \lim_{x \to 1} \frac{(x+1)(x-1)}{x - 1} = \lim_{x \to 1} (x + 1) = 1 + 1 = 2.$$

In this case, we are able to plug $x = 1$ into the expression for the limit to arrive at the same result.

Example. Let us compute

$$\lim_{x \to -1} \frac{\sqrt{x+10} - \sqrt{8-x}}{x^2 - 1}.$$

Let

$$f(x) = \frac{\sqrt{x+10} - \sqrt{8-x}}{x^2 - 1}.$$

Note that plugging $x = -1$ directly into the expression forces the denominator to be 0, and thus $f(-1)$ is undefined. We can try to rewrite $f(x)$ by *rationalizing* the numerator:

$$\begin{aligned}
\frac{\sqrt{x+10} - \sqrt{8-x}}{x^2 - 1} \cdot \frac{\sqrt{x+10} + \sqrt{8-x}}{\sqrt{x+10} + \sqrt{8-x}} &= \frac{(x+10) - (8-x)}{(x^2-1)\left(\sqrt{x+10} + \sqrt{8-x}\right)} \\
&= \frac{2x + 2}{(x^2-1)\left(\sqrt{x+10} + \sqrt{8-x}\right)} \\
&= \frac{2(x+1)}{(x+1)(x-1)\left(\sqrt{x+10} + \sqrt{8-x}\right)}.
\end{aligned}$$

We can thus proceed to compute the limit by simplifying our expression and directly substituting $x = -1$:

$$\begin{aligned}
\lim_{x \to -1} f(x) &= \lim_{x \to -1} \frac{2(x+1)}{(x+1)(x-1)\left(\sqrt{x+10} + \sqrt{8-x}\right)} \\
&= \lim_{x \to -1} \frac{2}{(x-1)\left(\sqrt{x+10} + \sqrt{8-x}\right)} \\
&= \frac{2}{(-1-1)\left(\sqrt{-1+10} + \sqrt{8-(-1)}\right)} \\
&= \frac{2}{(-2)\left(\sqrt{9} + \sqrt{9}\right)} = -\frac{1}{6}.
\end{aligned}$$

In some cases, we may also be interested in **limits at infinity**.

Definition. If $f(x)$ approaches the value of L by taking x to be sufficiently large, then we write

$$L = \lim_{x \to \infty} f(x),$$

and we say that the L is the **limit** of $f(x)$ as x approaches **infinity**. Similarly, if $f(x)$ approaches L by taking x to be a sufficiently large negative value, then we write

$$L = \lim_{x \to -\infty} f(x),$$

and we say that L is the **limit** of $f(x)$ as x approaches **negative infinity**.

Example. Let us compute

$$\lim_{x\to\infty} \frac{\sin x}{x}.$$

To investigate this limit, we can recall a key property of the sine function: the *range* of $\sin x$ is $[-1, 1]$; in particular,

$$-1 \leq \sin x \leq 1.$$

Note, then, dividing the inequalities throughout by x (> 0),

$$-\frac{1}{x} \leq \sin x \leq \frac{1}{x}.$$

This suggests that the limit of $\sin x$ when x approaches infinity must lie in between the limits of $-1/x$ and $1/x$:

$$\lim_{x\to\infty} -\frac{1}{x} \leq \lim_{x\to\infty} \sin x \leq \lim_{x\to\infty} \frac{1}{x}.$$

We know that both $-1/x$ and $1/x$ approach 0 as x gets larger, and so we can conclude that

$$0 \leq \lim_{x\to\infty} \sin x \leq 0 \quad \Longrightarrow \quad \lim_{x\to\infty} \sin x = 0.$$

The above example demonstrates an application of the **Squeeze Theorem**.

Theorem. *Suppose that $f(x) \leq g(x) \leq h(x)$ when x is near a. If,*

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L,$$

then

$$\lim_{x\to a} g(x) = L.$$

In other words, if a function $g(x)$ is 'squeezed' between $f(x)$ and $h(x)$ near a, and both $f(x)$ and $h(x)$ have the same limit at this point, then $g(x)$ must have the same limit as well.

Extra Notes on Limits

Limit Laws. We have a few rules on limits which we often assume and must thus be familiar with. Suppose we have two functions $f(x)$ and $g(x)$, and that the limits

$$\lim_{x\to a} f(x) \quad \text{and} \quad \lim_{x\to a} g(x)$$

exist. Then, we have:

1. $\lim\limits_{x\to a} [cf(x)] = c \lim\limits_{x\to a} f(x)$ for all constants c
2. $\lim\limits_{x\to a} [f(x) + g(x)] = \lim\limits_{x\to a} f(x) + \lim\limits_{x\to a} g(x)$
3. $\lim\limits_{x\to a} [f(x)\, g(x)] = \lim\limits_{x\to a} f(x) \lim\limits_{x\to a} g(x)$
4. $\lim\limits_{x\to a} \left[\dfrac{f(x)}{g(x)}\right] = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ if $\lim\limits_{x\to a} g(x) \neq 0$

One-Sided Limits. There may be some functions for which, approaching a desired point $x = a$ from the left (that is, with values of x less than a) may produce a different limit when compared to approaching $x = a$ from the right (with values of x greater than a). It is thus sometimes helpful to make the following distinction.

Definition. We write

$$L = \lim_{x\to a^-} f(x),$$

and say that L is the **left-hand limit** of $f(x)$ as x approaches a, if we can make the values of $f(x)$ close to L by taking $x$sufficiently close to a from the left, that is, with $x < a$. Similarly, we write

$$L = \lim_{x\to a^+} f(x),$$

and say that L is the **right-hand limit** of $f(x)$ as x approaches a, if we we can make the values of $f(x)$ close to L by taking $x$sufficiently close to a from the right, that is, with $x < a$.

It is thus clear that the following result is true:

$$L = \lim_{x\to a} f(x) \quad \text{if and only if} \quad L = \lim_{x\to a^-} f(x) \text{ and } L = \lim_{x\to a^+} f(x).$$